Preliminary Design and Trade Study of Supersonic Propulsion Systems

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Chapter 1

Problem Description

The renewed interest in commercial supersonic air transport has motivated the preliminary design of propulsion systems capable of efficient operation at high supersonic cruise conditions. In this study, a conceptual commercial aircraft designed to cruise at Mach 3.2 is considered, with the objective of completing a preliminary comparison between two propulsion system candidates. The primary emphasis of the project is the design and analysis of the propulsion system rather than full aircraft sizing or configuration.

A critical component of supersonic propulsion system performance is the inlet, whose function is to decelerate the freestream supersonic flow to subsonic conditions suitable for the engine while minimizing total pressure losses. Poor inlet pressure recovery directly reduces available thrust and overall cycle efficiency. As such, the inlet design strongly influences the feasibility of high-speed commercial propulsion systems.

The first phase of this project focuses on the preliminary design of a supersonic inlet for a fictional engine using classical compressible flow theory. The inlet is designed for a cruise Mach number of 3.2 and employs a multi-shock compression system consisting of three oblique shocks followed by a terminal normal shock. The inlet is optimized using the Oswatitsch equal-strength shock principle to maximize total pressure recovery across the inlet.

Chapter 2

Approach

The inlet design problem is approached using one-dimensional compressible flow theory and classical shock relations. For a supersonic inlet employing multiple oblique shocks followed by a normal shock, total pressure losses arise from irreversible entropy increases across each shock. Oswatitsch demonstrated that, for a given number of shocks and flight Mach number, the total pressure recovery is maximized when all oblique shocks are of equal strength [1].

Equal shock strength is achieved when the Mach number normal to each oblique shock is constant throughout the inlet:

$$M_1 \sin\beta_1 = M_2 \sin\beta_2 = \cdots = M_{n-1} \sin\beta_{n-1} \tag{2.1}$$

where M_i is the upstream Mach number of the i-th oblique shock and β_i is the corresponding shock angle.

For each oblique shock, the flow deflection angle θ_i and downstream Mach number are obtained from the standard oblique shock relations. The stagnation pressure loss across each oblique shock is calculated using the equivalent normal shock relation based on the normal Mach number component. The final normal shock is designed such that the upstream Mach number immediately before the shock is 1.3, as specified in the project requirements.

The system of equations is solved numerically using a root-finding approach. A single unknown parameter—the common normal Mach number for all oblique shocks—is iteratively adjusted until the Mach number upstream of the terminal normal shock matches the specified value. This approach ensures numerical robustness while strictly enforcing the Oswatitsch condition.

Chapter 3

Supersonic Inlet Design

3.1 Design Inputs

The inlet is designed for the following flight and design conditions:

- Freestream Mach number: $M_1 = 3.2$
- Ratio of specific heats: $\gamma = 1.4$
- Number of oblique shocks: 3
- Upstream Mach number before terminal normal shock: $M_n = 1.3$

3.2 Oblique Shock Solution

Application of the Oswatitsch equal-strength condition resulted in a common normal Mach number across all oblique shocks of:

$$M_n^* = 1.493$$

The resulting shock angles, flow deflection angles, and Mach number progression through the inlet are summarized in Table 3.1.

Table 3.1: Oblique shock properties for Oswatitsch-optimized inlet

Shock	M_i	β_i (deg)	θ_i (deg)	M_{i+1}
1	3.200	27.82	11.90	2.566
2	2.566	35.60	14.45	1.950
3	1.950	49.98	17.22	1.300

The results show a smooth and monotonic reduction in Mach number through the oblique shock system, while maintaining supersonic flow prior to the terminal normal shock.

3.3 Pressure Recovery

The stagnation pressure ratio across each oblique shock was found to be identical, consistent with the equal-strength shock condition:

$$\pi_1 = \pi_2 = \pi_3 = 0.9319$$

The stagnation pressure ratio across the terminal normal shock was computed as:

$$\pi_4 = 0.9794$$

The total inlet pressure recovery is therefore:

$$\pi_d = \prod_{i=1}^{4} \pi_i = 0.7926$$

This value is consistent with expected pressure recovery levels for Mach 3+ external-compression inlets and confirms the effectiveness of the Oswatitsch-optimized shock system.

3.4 Inlet Geometry

A two-dimensional ramp inlet geometry corresponding to the calculated deflection angles was generated to provide a visual representation of the inlet configuration. The inlet consists of three successive compression ramps producing the required flow turning prior to the terminal normal shock. A schematic of the inlet geometry is shown in Figure 3.1.

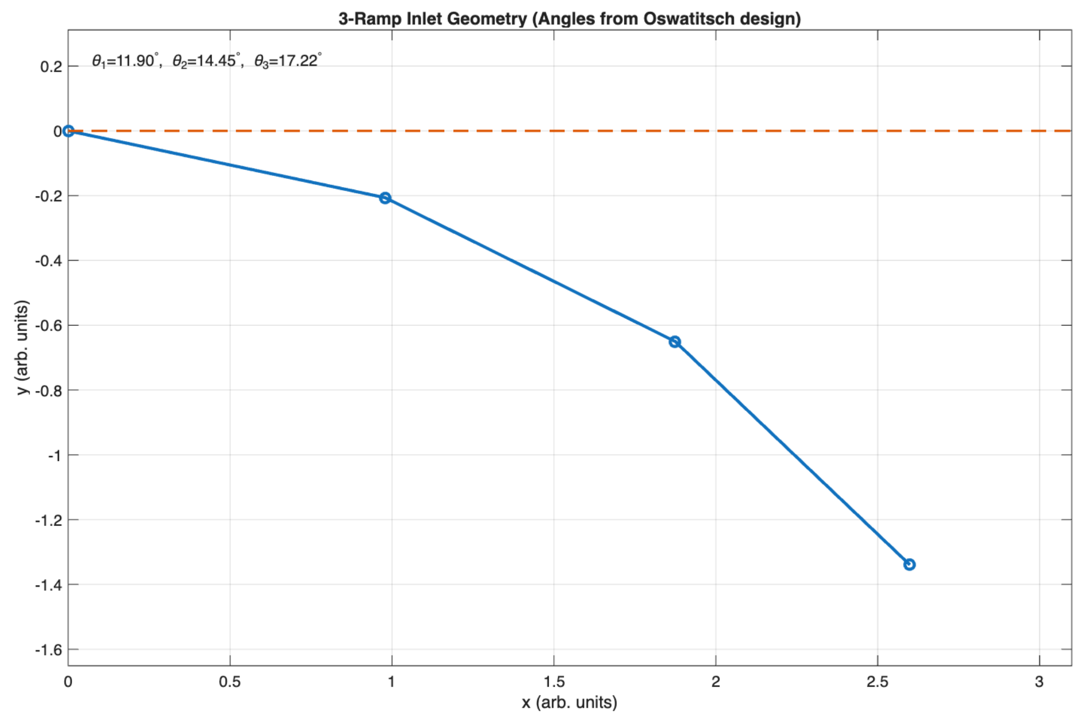

Figure 3.1: Three-ramp supersonic inlet geometry based on Oswatitsch-optimized deflection angles

Chapter 4

Parametric Cycle Analysis

A one-dimensional parametric cycle analysis was performed to evaluate the performance of a fictional turbojet engine designed specifically for sustained Mach 3.2 cruise. The analysis incorporated the inlet pressure recovery obtained from the Oswatitsch-optimized inlet design presented in Part I, with a total inlet pressure ratio of $\pi_d = 0.793$.

The cycle analysis examined the influence of compressor pressure ratio (π_c) and turbine inlet temperature (T_{t4}) on key performance metrics, including specific thrust and thrust-specific fuel consumption (TSFC). The flight condition was fixed at Mach 3.2 and an altitude of 15 km to represent the intended cruise condition.

4.1 Parametric Study Setup

The compressor pressure ratio was varied from $\pi_c = 8$ to 30, while turbine inlet temperature was varied from $T_{t4} = 1500$ K to 2000 K. Component efficiencies and pressure losses were selected based on typical values for preliminary turbojet analysis. Regions producing negative or non-physical thrust were excluded from the TSFC evaluation to ensure physically meaningful results.

4.2 Parametric Trends

The parametric results demonstrate that specific thrust decreases with increasing compressor pressure ratio at high supersonic flight speeds. This trend reflects the increasing compressor work requirement and diminishing benefit of higher pressure ratios at elevated inlet total temperatures. Conversely, specific thrust increases strongly with turbine inlet temperature, indicating that high T_{t4} values are required to overcome ram drag at Mach 3.2.

The TSFC results exhibit a trade-off between efficiency and thrust. Low T_{t4} and low π_c configurations minimize TSFC but fail to provide sufficient specific thrust for sustained cruise. Imposing a minimum specific thrust constraint shifts the optimal design toward higher turbine inlet temperatures and moderate compressor pressure ratios.

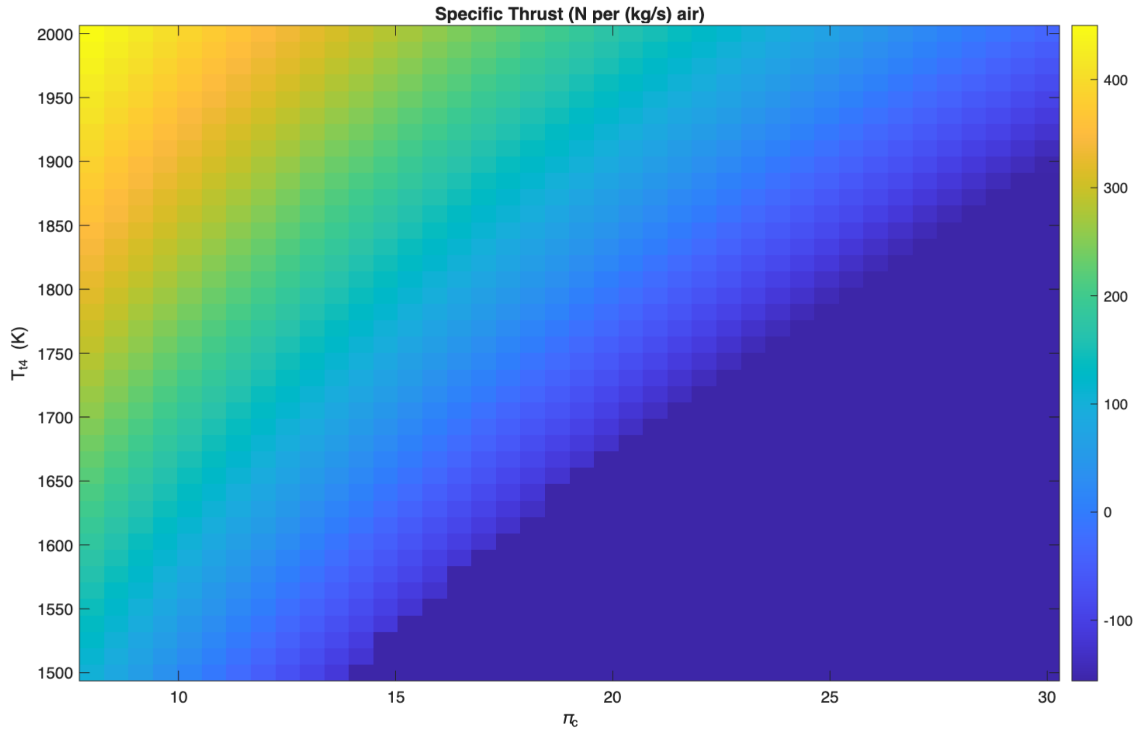

Figure 4.1: Specific thrust variation with compressor pressure ratio and turbine inlet temperature for the fictional turbojet at Mach 3.2.

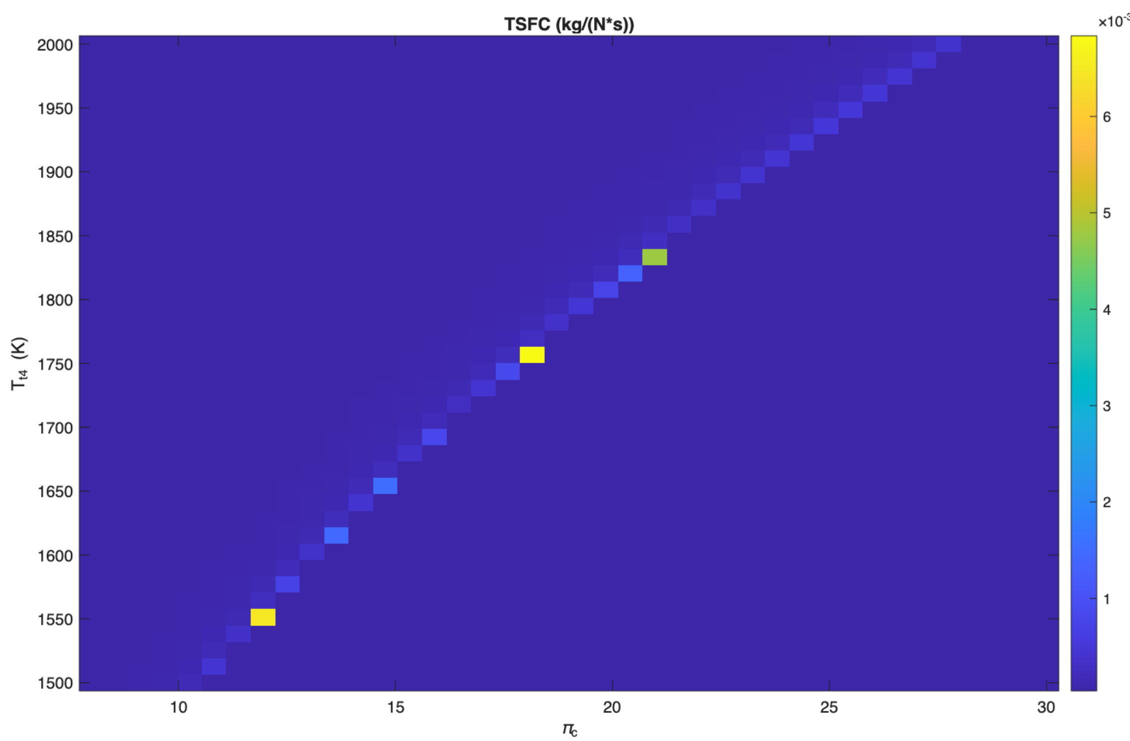

Figure 4.2: TSFC variation for physically feasible operating points of the fictional turbojet at Mach 3.2.

Chapter 5

Engine Trade Study

5.1 Fictional Engine Design Point

Based on the parametric analysis, a feasible fictional engine design point was selected by imposing a minimum specific thrust requirement of 250 N/(kg/s) to ensure adequate propulsion capability at Mach 3.2. The resulting design point is summarized in Table 5.1.

Table 5.1: Selected fictional engine design point at Mach 3.2

Parameter	Value
Compressor pressure ratio, π_c	10.26
Turbine inlet temperature, T_{t4}	2000 K
Specific thrust	390 N/(kg/s)
TSFC	4.04×10^{-5} kg/(N·s)

This design reflects the necessity of elevated turbine inlet temperature to achieve positive net thrust at high supersonic cruise conditions, while maintaining a moderate compressor pressure ratio to limit excessive compressor work.

5.2 Off-the-Shelf Engine Comparison

The Rolls-Royce/Snecma Olympus 593 engine, developed for the Concorde supersonic transport, was selected as an off-the-shelf benchmark for comparison. The Olympus 593 was designed for efficient cruise near Mach 2 and employed a higher compressor pressure ratio but significantly lower turbine inlet temperature relative to the fictional engine.

When evaluated at the Mach 3.2 mission condition using the same cycle framework, the Olympus 593 baseline configuration did not produce positive net thrust within the steady one-dimensional turbojet model. This result indicates that the original Olympus 593 cycle, optimized for lower supersonic speeds, is not suitable for sustained Mach 3.2 cruise without substantial modification.

5.3 Discussion

The comparison highlights the critical role of turbine inlet temperature and inlet pressure recovery in enabling high-Mach-number cruise. While the Olympus 593 achieves acceptable performance at Mach 2 through a combination of moderate turbine temperature and afterburner-assisted operation, its lower T_{t4} limits its ability to overcome ram drag at Mach 3.2.

In contrast, the fictional engine is explicitly designed around the Mach 3.2 mission requirement, incorporating a higher allowable turbine inlet temperature and an optimized inlet to maintain sufficient total pressure at the engine face. These design choices enable sustained supersonic cruise at the cost of increased thermal loading and higher fuel consumption, illustrating the fundamental trade-offs inherent in high-speed propulsion system design.

5.4 Summary Comparison

Table 5.2: Comparison of fictional engine and Olympus 593 at Mach 3.2

Parameter	Fictional Engine	Olympus 593
Design Mach number	3.2	2.0
Compressor pressure ratio	10.26	15.5
Turbine inlet temperature (K)	2000	1355
Inlet pressure recovery	0.793	0.793
Specific thrust	390 N/(kg/s)	Not feasible
TSFC	4.04×10^{-5}	Not feasible

Chapter 6

Conclusions

A preliminary propulsion system design and trade study was conducted for a commercial aircraft intended to cruise at Mach 3.2. An Oswatitsch-optimized supersonic inlet was successfully designed to maximize total pressure recovery, yielding an inlet pressure ratio of 0.793. This inlet formed the foundation for subsequent cycle analysis.

A parametric one-dimensional cycle analysis demonstrated that sustained Mach 3.2 cruise requires elevated turbine inlet temperatures and moderate compressor pressure ratios to maintain positive net thrust. A fictional turbojet engine designed specifically for the mission achieved a specific thrust of approximately 390 N/(kg/s) at a turbine inlet temperature of 2000 K.

Comparison with the Olympus 593 off-the-shelf engine revealed that, while the Olympus was well-suited for Mach 2 cruise, it could not sustain Mach 3.2 operation in its baseline configuration. This outcome emphasizes the importance of tailoring propulsion system design to specific mission requirements rather than extrapolating existing designs beyond their intended operating envelopes.

Overall, the study demonstrates that achieving commercial Mach 3.2 cruise necessitates significant advancements in turbine materials, cooling technology, and inlet performance, and underscores the value of mission-driven propulsion system design in preliminary aerospace engineering studies.

Bibliography

[1] K. Oswatitsch, “Pressure recovery for missiles with reaction propulsion at high supersonic speeds,” *NACA TM 1140*, 1947.

Appendix A

Design Code

All inlet design and cycle analysis calculations were performed using custom MATLAB scripts developed specifically for this project. The code is fully self-contained and may be executed by running the front-end scripts `run_inlet_design.m` and `run_cycle_analysis.m`. Supporting functions implement standard compressible flow relations and one-dimensional turbojet cycle equations.

Representative excerpts are shown below for reference.

A.1 Inlet Design Solver

```
oswatitsch_solver.m
oblique_shock.m
normal_shock.m
```

A.2 Cycle Analysis

```
cycle_analysis.m
parametric_study.m
engine_params_baseline.m
engine_params_olympus593.m
```

The full codebase is available in the project submission directory.